

Ting-Wei Chen

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OBJECTIVE

MEng student in Chemical Engineering seeking a Process Engineer role at TSMC. Experienced in transport modeling, materials analysis, and process simulation relevant to semiconductor manufacturing.

EDUCATION

Oregon State University — MEng, Chemical Engineering (Expected June 2026)

Yuan Ze University — B.E. Chemical Engineering & Materials Science (June 2023)

TECHNICAL SKILLS

- Modeling: COMSOL
- Engineering: Diffusion, reaction engineering, mass balance
- Materials: SEM, XRD
- Programming: Python, Excel

EXPERIENCE

Microreactor Modeling (CHE 540)

- Modeled transport and reaction behavior in a catalytic microreactor using convection–diffusion equations
- Simulated system performance under varying flow rates and boundary conditions to evaluate concentration profiles
- Applied transport modeling approaches relevant to semiconductor manufacturing processes
- Supported process optimization by analyzing transport behavior under different operating conditions

SOFC Research Project

- Conducted research on gas transport in porous materials using SEM and XRD
- Increased gas permeability by 20% and porosity by 30% through material optimization
- Evaluated transport behavior by testing multiple CMF-based material concentrations
- Generated insights to improve material performance and support engineering design decisions